MIL-E-24355B(SH)
16 April 1982
SUPERSEDING
MIL-E-24355A(SHIPS)
24 October 1975
(See 6.5)

MILITARY SPECIFICATION

ELECTRODES, WELDING, BARE, SOLID, NICKEL-MANGANESE-CHROMIUM-MOLYBDENUM ALLOY STEEL FOR PRODUCING HY-130 WELDMENTS FOR AS-WELDED APPLICATIONS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for bare steel electrodes for welding HY-130 steel by using mechanized (automatic) gas tungsten-arc (GTA) process, manual (semi-automatic) or mechanized (automatic) gas metal-arc (GMA) welding process with spray transfer type arc, or pulsed spray welding type arc.

1.2 Classification. Electrodes covered by this specification are classified on the basis of type, welding position, size, and base metal thickness to be welded as shown in table I, as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3439

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TABLE I. Classification.

Type	Base metal thickness to be welded	
MIL-140S-1	3/4-inch and thicker	
	Electrode size, diameter (inch)	Welding position
	0.035, 0.045	All
	1/16 (0.062) 3/32 (0.094)	Horizontal fillets and flat

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-O-925 - Oxygen, Technical, Gas and Liquid.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

MILITARY

MIL-P-116 - Preservation-Packaging; Methods of.
MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-A-18455 - Argon, Technical.
MIL-S-24371 - Steel Plate, Alloy, Structural, High Yield Strength (HY-130).
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

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MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-1601 - Method for 5/8 Inch Dynamic Tear Testing of Metallic Materials.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSEA 0900-LP-003-9000 - Radiographic Standards for Production-Repair Welds.
NAVSEA 0900-LP-005-5000 - Standard Procedures for Preproduction Testing Materials by the Explosion Bulge Test.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY (AWS)
A3.0 - AWS Definitions - Welding and Cutting.
B4.0 - Standard Methods for Mechanical Testing of Welds. (DoD adopted)
Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Qualification. Electrodes furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening bids (see 4.3 and 6.3).

3.2 Definitions. For the purpose of this specification the welding terms and definitions contained in AWS A3.0 and 3.2.1 shall apply.

3.2.1 Contractor. The contractor is defined as the actual processor of the welding electrodes, engaged in the final cleaning, spooling, marking and packaged operations.

3.3 Alloy identity. Each end of rod or wire to be spliced during processing shall be tested for alloy identity (see 4.7.1), except when splicing is done to repair a wire break without removing the wire from the process line.

3.3.1 Both ends of each coil drawn to final size shall be tested for alloy identity (see 4.7.1) before rewinding and spooling.

3.3.2 When specified (see 6.2), each electrode shall be tested for alloy identity in accordance with 4.7.1 after final marking. Identification marking of each spool tested after final marking and meeting the acceptance criteria of 4.7.1 shall include the words "ALLOY TESTED".

3.4 Finish. Electrodes shall have a clean, smooth, and bright finish that will not rust in the original unopened container stored under roof and on dry platforms. Electrodes shall be free from slivers, depressions, scratches, and scale or foreign matter that would adversely affect the welding characteristics, the operation of the equipment, or the properties of the deposited metal.

3.5 Hydrogen content. Electrode shall be processed and packaged so that the hydrogen content is less than 3.0 parts per million (p/m) when sampled and analyzed as described in 4.7.3.

3.6 Dimensional tolerances. The diameter of the electrode shall not vary from the nominal diameter by more than plus or minus 0.001 inch.

3.7 Winding. Spools shall be level and layer wound and shall uncoil freely without tangling or bumping. Each layer shall be one diameter thick without cross-overs or wedging.

3.8 Cast and helix. The cast and helix of spooled electrode shall be such as to have imparted to the electrode a curve such that a specimen sufficient in length to form one loop when cut therefrom and placed on a flat surface shall form an unrestrained circle within diametrical limits specified in table II.

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TABLE II. Cast and helix.

Form (spools)	Cast (inches)		Helix (inches)[1/] (maximum)
	Minimum	Maximum	
3a	5	10	1/2
3b (0.030 and smaller)	12	30	-----
3b (0.035 and larger)	15	36	1-1/2
3f	10	30	1

[1/] Vertical separation between overlapped ends.

3.9 Chemical composition. Electrodes shall conform to the chemical composition specified in table III.

TABLE III. Chemical composition (percent by weight).[1/]

Elements	MIL-140S-1
Carbon	0.12
Manganese	1.50/2.00
Phosphorus	0.010
Sulfur	.010
Silicon	.30/0.50
Copper	.15
Nickel	1.95/3.10
Chromium	0.65/1.05
Molybdenum	.40/1.00
Vanadium	.04
Titanium	.04
Zirconium	.04
Aluminum	.04

[1/] Single values are maximum.

3.10 Welding characteristics. During welding, electrodes shall operate smoothly without interruption or excessive spatter, and shall produce weld beads of uniform appearance and form.

3.11 Groove weld properties.

3.11.1 Soundness. Electrodes shall be capable of depositing groove welds which are free from slag entrapments, weld metal cracking, and porosity in excess of grade I of NAVSEA 0900-LP-003-9000. Radiographic indications having a maximum length of 1/8 inch may be evaluated as porosity.

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3.11.2 Mechanical properties. When the electrode is deposited and the weld metal tested in accordance with 4.7.4 through 4.7.4.7, the as-welded mechanical properties shall be those specified in table IV.

TABLE IV. As-welded mechanical property requirements. 1/

Type	Test weldment thickness (inch)	Yield strength lb/in^2 2/	Elongation percent in 2 inches minimum	Transverse side bend	Dynamic tear			Explosion bulge
					Test specimen size inch	Energy ft/lbs minimum	Temperature °F	
MIL-140S-1	1	135,000 to 150,000	14	3/	5/8	475	30	4/

1/ The ultimate tensile strength and percentage reduction of area shall be recorded for information only.

2/ The yield strength shall be measured at 0.2 percent offset by using a class B extensometer or better. For the one inch thick weldment, the yield strength shall be the average of two specimens with a minimum single value of 135,000 lb/in^2.

3/ The convex surface of the specimen after bending shall have no visual cracks exceeding 1/8 inch. The corners of the specimen shall have no visual cracks exceeding 3/16 inch. The bend radius shall be 3 times the specimen thickness (3t) and tests shall be conducted as specified in 4.7.4.7.

4/ The explosion bulge test shall be conducted as specified in 4.7.5. The criteria of performance shall be in accordance with the following:

(a) Two crack-starter weldments.
- (1) First shot - Crack-starter bead shall crack.
  Percent reduction in thickness obtained for information only.
  No piece shall be thrown out of weldments.
  No through-thickness cracks shall be present.
  No cracks shall extend into hold-down area.*
- (2) Second shot - Percent reduction in thickness obtained for information only.
  No piece shall be thrown out of weldment.
  Through-thickness cracks are acceptable.
  No cracks shall extend into hold-down area.*

(b) Two bulge weldments.
- (1) First shot - Percent reduction in thickness obtained for information only.
  No piece shall be thrown out of weldment.
  No through-thickness cracks shall be present.
  No crack shall extend into hold-down area.*
- (2) Second shot - Percent reduction in thickness obtained for information, 5 percent reduction is expected.
  No piece shall be thrown out of weldment.
  No through-thickness cracks shall be present.
  No crack shall extend into hold-down area.*

See asterisk on next page.

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In the event that 7 percent reduction in plate thickness is obtained on both sides of the weld or 7-1/2 percent reduction in plate thickness is obtained on one side of the weld the performance is considered satisfactory and the third shot shall not be made provided the following conditions are met:

No piece shall be thrown out of weldment.
Through-thickness cracks are acceptable.
No cracks shall extend into the hold-down area.*

(3) Third shot - Percent reduction in thickness obtained for information, 7 percent reduction is expected.
No piece shall be thrown out of weldment.
Through-thickness cracks are acceptable.
No cracks shall extend into hold-down area.*

*Crack shall remain within 12-1/2-inch radius measured from center of weldment.

3.12 Form 3a, 3b, and 3f spools. Form 3a and 3b spools shall conform to MIL-W-10430. Form 3f spools shall conform to figure 1. Electrodes shall be wound on spools in the forms, sizes, and weight combinations shown in table V, as specified (see 6.2).

TABLE V. Form, size, and weight.

Form (spools)	Electrode size, diameter inch 1/	Net weight of electrode (pounds) 1/
3a	0.035, 0.045	1-1/2 or 2-1/2
3b	0.035, 0.045, 1/16 (0.062), 3/32 (0.094)	25 or 35
3f	0.035, 0.045	8, 10, or 12

1/ The net weight of the electrode shall not vary from the specified weight by more than plus or minus 10 percent.

3.13 Stability. The stability of the electrodes and of the packaging shall be such that electrodes after receipt from the manufacturer will comply with this specification after storage in original unopened containers under roof and on dry platforms for a warranty period of one year. This requirement does not establish the shelf life of the electrode but only the time period in which receipt inspection for acceptance applies.

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3.14 Electrode identification. The specification number, MIL-type, size, and lot control number shall appear on the flange of the spool so as to be visible when the spool is positioned on the machine for welding. The identification shall be placed in a manner that is not readily removable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification tests will be authorized only to contractors of welding electrodes. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Qualification shall consist of the examination and tests specified in table VI.

4.3.1 Special instructions. When applying for test authorization, or after tests have been authorized, the contractor shall furnish the following information to NAVSEA. (NOTE: This information, together with test results obtained with the electrode sample, shall form a part of the qualification test; all information will be held in confidence by the Government.)

(a) MIL-type and size under which approval is desired.
(b) Composition of wire in terms of nominal percentages for each constituent.
(c) Shielding gas composition.
(d) Brand name of electrode.
(e) Cleaning method.

4.3.2 Samples for qualification tests. An electrode sample of MIL-type and size for which qualification is desired shall be selected at one time and submitted for qualification testing. Qualification will be granted only for the MIL-type and size which has been submitted and which has successfully passed the qualification tests. The number of spools of electrode selected shall be a sufficient quantity to perform the required tests. The samples shall be selected in the presence of the Government representative.

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4.3.3 Summary of qualification tests. Table VI summarizes the examination and tests required under this specification for qualification testing of electrodes.

TABLE VI. Summary of examination and tests required for qualification.

Type	Form	Size[1]	Visual and dimensional	Winding	Cast and helix	Alloy identity	Chemical	Hydrogen[2]	Radiographic	Tensile	Bend	Dynamic tear	Explosion bulge[3]	Examination and test procedure	Requirement
MIL-140S-1	3b	All	X	---	---	---	---	---	---	---	---	---	---	4.6.1 4.6.2	3.4, 3.6 and table V
			---	X	---	---	---	---	---	---	---	---	---	4.6.4	3.7
			---	---	X	---	---	---	---	---	---	---	---	4.6.3	3.8
			---	---	---	X	---	---	---	---	---	---	---	4.7.1	3.3
			---	---	---	---	X	---	---	---	---	---	---	4.7.2	Table III
			---	---	---	---	---	X	---	---	---	---	---	4.7.3	3.5
			---	---	---	---	---	---	X	---	---	---	---	4.7.4.6	3.11.1
			---	---	---	---	---	---	---	X	X	---	---	4.7.4.7.3	Table IV
			---	---	---	---	---	---	---	---	---	X	---	4.7.4.7.4	Table IV
			---	---	---	---	---	---	---	---	---	---	X	4.7.5	Table IV

[1] Groove joints (figures 2 and 3 and explosion bulge weldments) shall be welded in the flat position for 0.062-inch and 0.094 diameter electrodes and in the vertical position for 0.035 and 0.045-inch diameter electrodes. The electrode sample selected in accordance with 4.3.2 shall be large enough to make all welds required by figures 2 and 3 and the explosion bulge test.

[2] Sample taken in accordance with 4.4.7.

[3] If the results of the mechanical tests of NAVSEA 0900-LP-005-5000 indicate that the weld metal properties of a smaller size electrode form the same welding position are equivalent to those of a larger size electrode of the same formulation previously qualified, the explosion bulge test shall be omitted. Evaluation of the weld metal properties by NAVSEA shall be used on the 5/8-inch dynamic tear impact energy curve equivalency at the same strength level.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table VII for each lot of electrodes as defined in 4.4.1.

TABLE VII. Summary of examination and tests required for quality conformance inspection.

Type	Form	Size 1/	Visual and dimensional	Winding	Cast and helix	Alloy identity	Chemical	Hydrogen 2/	Radio-graphic	Tensile	Dynamic tear	Examination and test procedure	Requirement
MIL-140S-1	All	All	X	---	---	---	---	---	---	---	---	4.6.1 4.6.2	3.4, 3.6 and table V
			---	X	---	---	---	---	---	---	---	4.6.4	3.7
			---	---	X	---	---	---	---	---	---	4.6.3	3.8
			---	---	---	X	---	---	---	---	---	4.7.1	3.3
			---	---	---	---	X	---	---	---	---	4.7.2	Table III
			---	---	---	---	---	X	---	---	---	4.7.3	3.5
			---	---	---	---	---	---	X	---	---	4.7.4.6	3.11.1
			---	---	---	---	---	---	---	X	---	4.7.4.7.3	Table IV
			---	---	---	---	---	---	---	---	X	4.7.4.7.4	Table IV

1/ Groove joints (figures 2 and 3 and explosion bulge weldments) shall be welded in the flat position for 0.062 and 0.094 inch diameter electrodes and in the vertical position for 0.035 and 0.045-inch diameter electrodes. The electrode sample selected in accordance with 4.3.2 shall be large enough to make all welds required by figures 2 and 3 and the explosion bulge test.

2/ Sample taken in accordance with 4.4.7.

4.4.1 Sampling for quality conformance inspection.

4.4.1.1 Lot definition.

4.4.1.1.1 All examination and tests except the hydrogen analysis. Electrodes, of the same type and size, produced from the same heat of steel shall be considered a lot.

4.4.1.1.2 Hydrogen analysis. Electrodes, of the same type and size, produced from the same heat of steel and processed into electrodes during a single production run shall be considered a lot. (A single production run is considered to be the production of one type and size electrode without interruption for the production of another type or size electrode on the same equipment and process lines.)

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4.4.2 Sampling for inspection of form and filled containers. Unit packages, spools, and shipping containers shall be sampled and inspected in accordance with MIL-W-10430 and MIL-P-116.

4.4.3 Sampling for visual and dimensional examination of electrode. Sample electrode shall be selected either from the production line immediately prior to packaging (after spooling) or from filled unit packages. If selected from the production line, the total sample shall be in accordance with table VIII and the electrodes shall be selected throughout the "run" so that all parts of the "run" are represented. If selection is made after the packaging operation, the total sample shall be in accordance with table VIII. Sufficient electrodes to form one complete circle shall be taken from each sample spool.

4.4.3.1 Any sample electrode which contains material not conforming to 4.6.1 shall be counted as defective. The number of defective sample electrodes shall not exceed the acceptance number in table VIII.

TABLE VIII. Sampling for visual and dimensional examination.

Lot size in pounds	Spool forms					
	3a		3f		3b	
	Sample size	Acceptance number	Sample size	Acceptance number	Sample size	Acceptance number
Up to 25	5	0	2	0	All	--
26 to 50	7	0	3	0	All	--
51 to 180	10	0	5	0	2	0
181 to 300	15	1	8	0	3	0
301 to 500	25	1	13	0	5	0
501 to 800	35	2	18	1	7	0
801 to 1300	50	3	25	1	10	0
1301 to 3200	75	4	35	2	15	1

4.4.4 Sampling for cast and helix tests. Specimens for the cast and helix tests (see 4.6.3) shall be cut from spools selected in accordance with 4.4.3 and table VIII.

4.4.5 Sampling for chemical analysis. One specimen shall be selected from the lot (see 4.4.1.1.1) and chemically analyzed as specified in 4.7.2 to verify conformance to the chemical composition specified in 3.9.

4.4.6 Sampling for weld tests. Sufficient quantity of electrodes shall be selected from the lot (see 4.4.1.1.1) for use in preparation of welds required herein (see 4.7.4).

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4.4.7 Sampling for hydrogen analysis. One specimen shall be selected from the lot (see 4.4.1.1.2) and analyzed for hydrogen as described in 4.7.3. Length of sample shall be as shown in table IX. Each sample for hydrogen analysis shall be obtained by cutting about a 10-gram piece of electrode from a location at least one layer inside the coil. The electrode shall be handled only by using clean white gloves and shall be stored in a hermetically sealed container during the period between sampling and actual analysis. The sample shall not be contaminated by foreign matter or moisture or both. The sample shall be analyzed in the as-received condition with no prior surface preparation such as solvent cleaning or abrasion.

TABLE IX. Length of 10-gram sample.

Electrode size, diameter inches	Length in inches
0.035	80
.045	48
.062	26
.094	11

4.6 Visual and dimensional examination.

4.6.1 Visual examination. Each of the electrode samples, selected in accordance with table VIII, shall be examined for conformance with 3.4. During welding of the test plates, the feed performance of the electrode shall be observed for conformance with 3.4 and 3.7.

4.6.2 Dimensional examination. The diameter of the electrode sample shall be measured for conformance with table V and 3.6.

4.6.3 Cast and helix. The diameter and free height of an unrestrained ring sample of the electrode shall be measured for conformance with 3.8. If any sample fails, the test shall be performed on every spool of the lot (see 4.4.1.1.1) and all spools that fail shall be rejected.

4.6.4 Winding. The performance of the spooled electrode sample selected in accordance with 4.4.6 shall be observed during the welding specified in 4.7.4 for conformance to 3.7.

4.7 Test procedures.

4.7.1 Alloy identity.

4.7.1.1 Procedure. The alloy identity test method may include chemical analysis, metal sorting devices, or other approved methods or a combination of methods. The test method shall be submitted to NAVSEA for approval.

4.7.1.2 Acceptance criteria. If the test demonstrates that the material is not of the type specified, the material shall be rejected.

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4.7.2 Chemical composition of the electrode. The chemical composition of the electrode shall be determined in accordance with FED-STD-151.

4.7.3 Hydrogen analysis. The hydrogen content of the electrode shall be determined by using the hot-vacuum-extraction method in which the sample is outgassed by heating in the range 950°C to 1000°C at a pressure of 10^{-4} torr, or lower. The intert carrier gas extraction method at 1000°C to 1100°C may be used as an alternate hydrogen extraction method. In case of a dispute, hydrogen extraction by the hot-vacuum method shall be the basis for acceptance. The extracted gases shall be analyzed for hydrogen by one of the following techniques:

(a) Gas chromatography.
(b) Pressure measurement before and after catalytic oxidization of hydrogen to water and removal of the water.
(c) Pressure measurement before and after diffusion of hydrogen through a palladium membrane.
(d) Mass spectrometry.

In case of dispute, hydrogen analysis by mass spectrometry shall be the basis for acceptance.

4.7.3.1 Lot reprocessing. Any lot (see 4.4.1.1.2) that is rejected because of high hydrogen content may be reprocessed and tested a total of two times. Failure to meet the requirements of this specification after the second reprocessing shall cause final permanent rejection of the lot of electrodes. All other tests shall be performed with electrodes sampled after the last reprocessing operation.

4.7.4 Weld test groove joint, (see figures 2 and 3), shall be welded for qualification testing or quality conformance inspection to provide specimens for the radiographic and mechanical property tests required by table VI or table VII. When used for qualification testing, the test weldment (see figure 2), shall be extended to provide for the two transverse side bend test specimens required by table VI. The transverse side bend test specimens shall be taken next to the discard ends. At the option of the manufacturer, the test weldment (see figure 3), may be extended to provide for two extra 5/8-inch dynamic tear test coupons for retest, if needed.

4.7.4.1 Equipment. Semi-automatic or automatic gas metal-arc (GMA) welding equipment with single operator power sources shall be used. The equipment shall be suitable for spray transfer in the flat position and pulsed spray transfer in the vertical position.

4.7.4.2 Shielding gas. A mixture of argon and oxygen that consists of nominally 1.0 to 2.0 percent oxygen and the remainder argon shall be used for flat position welding. The electrode contractor shall recommend the exact gas composition for vertical welding. Argon and oxygen gases shall be in accordance with MIL-A-18455 and BB-O-925, respectively.

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4.7.4.3 Welding conditions. The welding conditions, other than those specified below, shall be determined by the electrode contractor. The process shall use direct current, reverse polarity; and the settings shall provide stable arc characteristics, adequate penetration, and suitable wetting.

4.7.4.4 Base material. The base-plate material used in preparation of test joints shall be HY-130 steel in accordance with MIL-S-24371.

4.7.4.5 Welding technique. Procedures for making qualification and quality conformance test welds shall conform to the following:

(a) A low cooling rate and a high cooling rate weldment shall be fabricated for each size electrode in 1-inch plate.
(b) The flat position with spray transfer shall be used for the 1/16 (0.062) inch diameter electrode and the vertical position with pulsed spray welding shall be used for the 0.035 and 0.045-inch diameter electrodes.
(c) Cooling rate parameters for figures 2 and 3 shall be in accordance with table X.
(d) Should welding be interrupted, the assembly shall be allowed to cool in still air at room temperature, and prior to resumption of welding, the assembly shall be reheated to the same preheat temperature as was used at the beginning of the test required by (a) or (c) above.
(e) After welding is completed, the weldment shall be allowed to cool in the air, and no postweld heat treatment shall be employed.

TABLE X. Cooling rate parameters

Type	Nominal weldment plate thickness (inch)	Cooling rate	Figure number	Preheat and interpass temperature (°F)	Heat input (joules per inch)
MIL-140S-1	1	Low	2	275 to 300	35,000 to 40,000
		High	3	150 to 175	30,000 to 35,000

4.7.4.6 Radiographic inspection. When 48 hours has elapsed after completion of the welding, the weld reinforcement and backing strips shall be removed flush with the base-plate on both surfaces. The welds of figures 2 and 3 shall be inspected radiographically in accordance with level 2-2T of MIL-STD-271 for compliance with the requirements of 3.11.1.

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4.7.4.7 Mechanical tests. Side bend, when required, and tensile test specimens shall be machined and tested in accordance with ANSI/AWS B4.0. The 5/8-inch dynamic tear test specimens shall be machined and tested in accordance with MIL-STD-1601 for compliance with the requirements of table IV. Two specimens are required for the initial test of MIL-STD-1601.

4.7.4.7.1 Removal of test coupons. Flame cutting shall not be used within 1/2 inch of the faces of the weld for the removal of test coupons.

4.7.4.7.2 Specimen finishing. Only sawing or machining shall be used for specimen finishing.

4.7.4.7.3 Side bend and tensile test. The transverse side bend test coupons required for qualification testing only shall be removed from the weldment of figure 2 next to the discard ends. The tensile test coupons shall be removed from the weldment of figure 2 as shown. The transverse side bend, when required, and tensile test specimens shall be finished and tested in accordance with ANSI/AWS B4.0.

4.7.4.7.4 Dynamic tear test. The 5/8-inch dynamic tear test coupons shall be removed from the weldment of figure 3 as shown. The test specimens shall be finished and tested in accordance with MIL-STD-1601 for compliance with the requirements of table IV.

4.7.4.8 Unsatisfactory test results. If the results of the first radiographic examination, bend, dynamic tear, or tensile tests are determined to be unsatisfactory, one retest involving twice the number of specimens originally required may be permitted. The results of all retests shall be satisfactory for lot acceptance.

4.7.5 Explosion bulge test. The explosion bulge test shall be conducted for MIL-140S-1 in accordance with NAVSEA 0900-LP-005-5000 using the following procedures:

(a) The base-plate material shall be HY-130 steel in accordance with MIL-S-24371.
(b) Direct current, reverse polarity shall be used.
(c) The preheat and interpass temperature shall be 150°F to 175°F and the heat input shall be 30,000 to 35,000 joules per inch.
(d) The flat position with spray transfer shall be used for the 1/16 (0.062) inch diameter electrode and the vertical position with pulsed spray welding shall be used for the 0.035 and 0.045-inch diameter electrodes.
(e) Should welding be interrupted, the assembly shall be allowed to cool in still air at room temperature, and prior to resumption of welding, the assembly shall be reheated to a temperature range of 150°F to 175°F.
(f) After welding is completed, the weldment shall be allowed to cool in the air, and no postweld heat treatments shall be employed.

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(g) The crack-starter and explosion bulge weldments shall be tested with the weld reinforcement in place.
(h) Explosive charge and standoff conditions for the crack starter plates shall be such that a smooth, unwelded plate of equivalent yield strength would attain a minimum through-thickness reduction of 5 percent at the center of the bulge after two shots.

4.8 Certificate of inspection. A certificate of quality conformance inspection of the tests specified in table VII containing actual quantitative tests and inspection results including the gas composition and range of amperage employed for each accepted lot included in a particluar shipment shall be furnished to the consignee with the shipment. The minimum quality conformance inspection result data required shall be as listed on figure 4 and shall be submitted on the form shown.

4.9 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 Spools. Spools shall be in accordance with MIL-W-10430 and figure 1 herein as required for the forms specified (see 3.12).

5.2 Preservation-packaging. Preservation-packaging shall be level A as specified in 5.2.1.

5.2.1 Level A. Electrodes, in the forms specified (see 3.12), shall be individually preserved-packaged in accordance with method IIe of MIL-P-116.

5.2.1.1 Intermediate packaging. Intermediate packaging shall be applied to form 3a and form 3f unit packages as follows:

Form 3a - 20 packages per container.
Form 3f - 8 pound spool - 4 packages per container.
Form 3f - 12 pound spool - 2 packages per container.

Intermediate containers shall conform to PPP-B-636, class weather-resistant, with closure and waterproofing conforming to method V of the appendix to the box specification.

5.2.1.2 Wrapping. Form 3b packages shall be overwrapped, to protect tne method IIe barrier from abrasion, with barrier material conforming to PPP-B-1055. Wrapping shall be secured with pressure sensitive tape.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2) in accordance with MIL-W-10430. Palletization, when required, shall conform to industry normal retail or wholesale practices.

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5.4 Marking. In addition to the special marking required herein or by the contract or order (see 6.2), interior packages, exterior shipping containers and palletized loads, shall be marked in accordance with MIL-W-10430 and shall include the method II precautionary label on all method II packages.

5.4.1 Special marking.

5.4.1.1 Spools. In addition to the identification marking reqiured (see 3.14), each spool shall be marked with the contractor's or distributor's name and brand or type designation. In addition, each spool and all unit packages shall be marked or labeled with the following or equivalent.

WARNING: Protect yourself and others. Read and understand this label.

FUMES and GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

- Read and understand the manufacturer's instructions and your employer's safety practices.
- Keep your head out of the fumes.
- Use enough ventilation or exhaust at the arc, or both to keep fumes and gases from your breathing zone, and the general area.
- Wear correct eye, ear and body protection.
- Do not touch live electrical parts.
- See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society, 2501 N.W. 7th St., Miami, FL 33125; OSHA Safety and Health Standards, 29 CFR 1910, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

DO NOT REMOVE THIS LABEL

6. NOTES

6.1 Intended use. This specification is intended to cover nickel-manganese-chromimum-molybdenum alloy steel solid bare welding electrodes capable of depositing weld metal, that in the as-welded condition meets the mechanical properties specified herein; when welded on HY-130 steel with the GMA process using direct current, reverse polarity and the techniques specified herein. The primary use of this electrode is for welding of HY-130 steel for applications in the as-welded condition. See table I for base metal thickness to be welded.

6.1.1 Type MIL-140S-1. This type is intended for welding HY-130 steel 3/4-inch and greater in thickness.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size (see 1.2).
(c) Whether alloy identity testing of each electrode after final marking is required (see 3.3.2).
(d) Form, size, and weight required (see 3.12).
(e) Levels of packing required (see 5.3).
(f) Special marking required (see 5.4).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL-24355 whether or not such products have actually been so listed by that date. The attention of the contractors is called to thee requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order thay they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

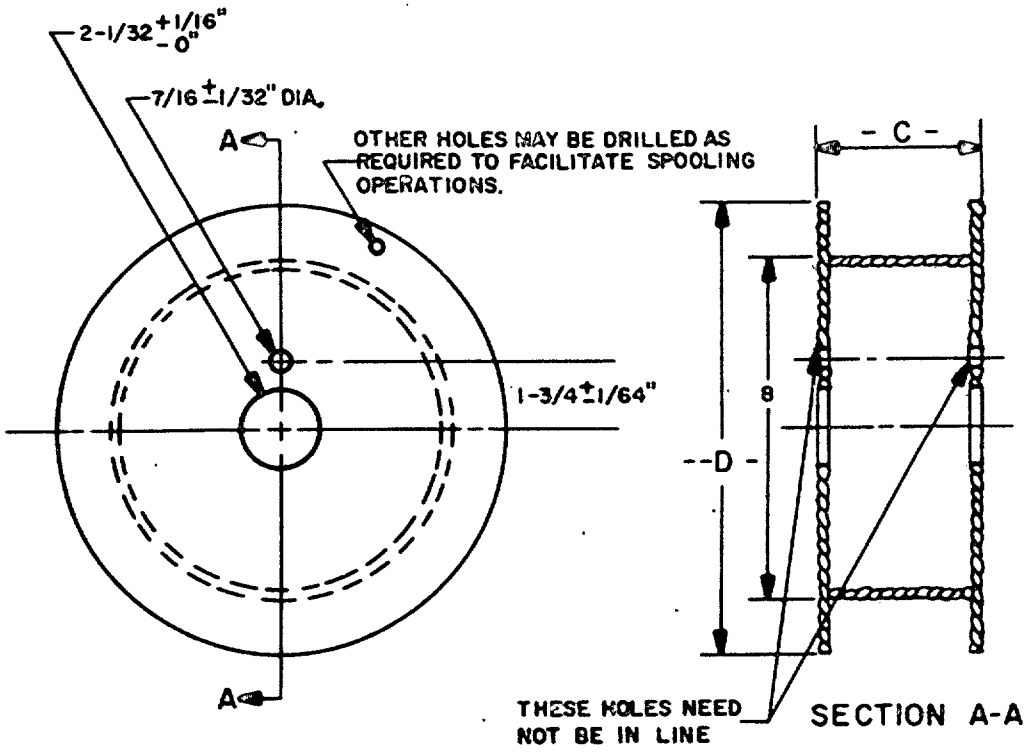
6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Electrode contractors may make arrangements to have 5/8-inch dynamic tear tests conducted by contacting the following: Commander, Mare Island Naval Shipyard (Code M138), Vallejo, CA 94592; Commander, Portsmouth Naval Shipyard (Code 374), Portsmouth, NH 03801 or Officer-in-Charge, Annapolis Laboratory, David W. Taylor Research and Development Center (Code 2821), Annapolis, MD 21402.

6.5 Changes from previus issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3439-N344)

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SH 10885

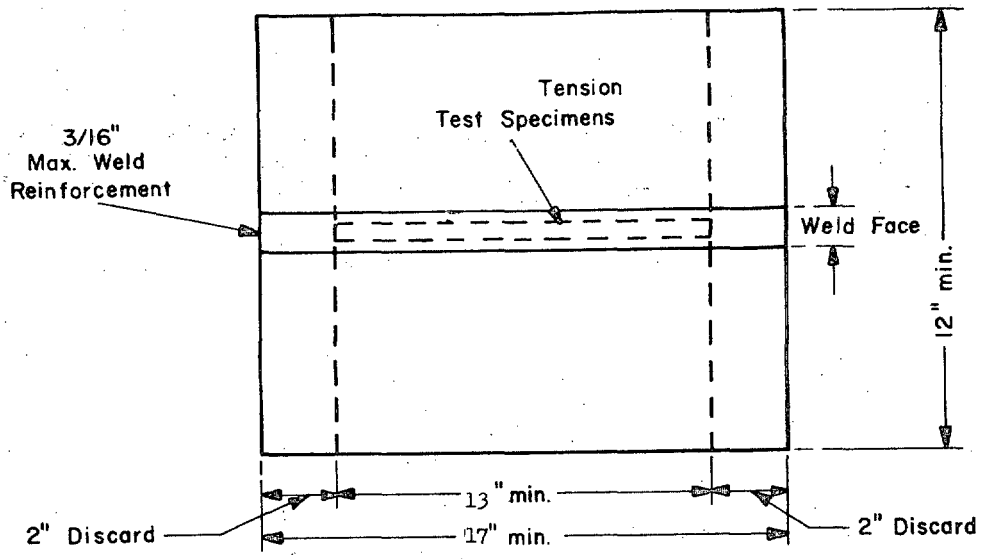
Form	B	Dimensions - inches	
		C	D (maximum)
3f	Note 2	2-5/32 + 1/32	8

NOTES:

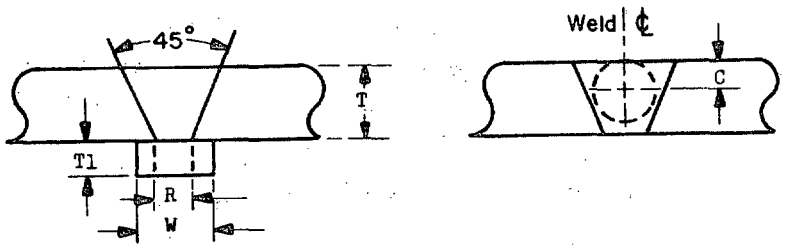
1. Flange and barrel material shall be nonconductive, and of such construction to carry the net weight of specified electrode and retain stability under handling, shipment, storage, and use. Spools shall be all plastic or constructed of hardboard flanges and cardboard barrels.
2. Dimension B shall be such as to permit proper feeding of electrode.
3. Flanges shall not be out of parallel more than 3/16 inch. Flanges shall not tee in.
4. The ends of the electrode shall be securely fastened to the hub and either flange by any means that permit accessibility to both ends, while satisfactorily retaining the electrode on the spool until used, and moreover, does not interfere with unspooling operation during use.

FIGURE 1. Details for form 3f spools.

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Layout of Test Weldment



Joint Geometry

Specimen Location

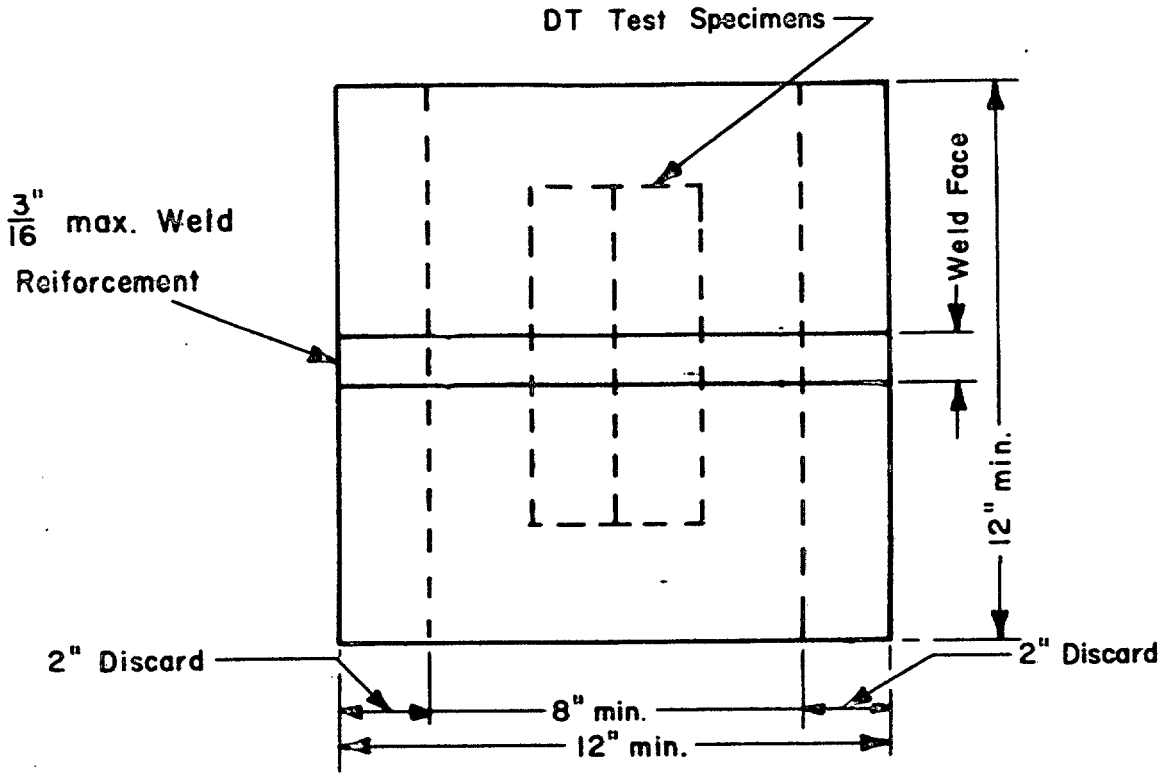
SH 10886

Type	Nominal[1/] thickness (T) (Inch)	Nominal[1/] thickness (T_1) (Inch)	Maximum width (W) (Inch)	Minimum opening (R) (Inch)	Dimension (C) (Inch)	Number of specimens	Diameter of tension specimen within gauge length (Inch)
MIL-140S-1	1	1/2	1	1/2	3/8	2	0.500 ± 0.010

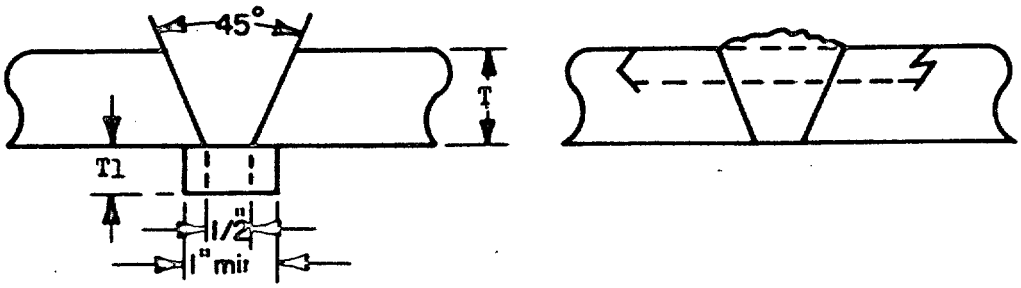
[1/] Tolerance on plate thickness shall be that specified in MIL-S-24371.

FIGURE 2. Low cooling rate test weldments.

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Layout of Test Weldment



SH 10887

Joint Geometry

Specimen Location

Type	Nominal[1/] thickness (T) (Inch)	Nominal[1/] thickness (T_1) (Inch)	Minimum width (W) (Inch)	Maximum opening (R) (Inch)	Number of specimens	Test specimen size (Inch)
MIL-140S-1	1	1/2	1	1/2	2	5/8

[1/] Tolerance on plate thickness shall be that specified in MIL-S-24371.

FIGURE 3. High cooling rate test weldment.

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Contractor or Distributor _____

Address _____

Date _____

Specification _____

Type _____

Size _____

Lot No. _____

Chemical Analysis

Carbon _____

Manganese _____

Phosphorus _____

Sulfur _____

Silicon _____

Copper _____

Nickel _____

Chromium _____

Molybdenum _____

Vanadium _____

Titanium _____

Zirconium _____

Aluminum _____

Hydrogen _____

Radiography Results _____

Visual & Dimensional _____

Cast & Helix _____

Customer's Name _____

Customer's Order No. _____

Mechanical Test Results

High Cooling Rate Test _____

5/8 DT Test at $30^{\circ}F$ (ft/lbs) _____

Low Cooling Rate Test _____

Yield Strength (lb/in^2) _____

Percent Elongation _____

Welding Procedure

High Cooling Rate

Process _____

Shielding Gas _____

Welding Current _____

Weld Energy Input _____

Welding Position _____

Preheat/Interpass Temp _____

Low Cooling Rate

Process _____

Shielding Gas _____

Welding Current _____

Weld Energy Input _____

Welding Position _____

Preheat/Interpass Temp _____

FIGURE 4. Certification of quality conformance inspection.

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We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTE: This form is applicable to the quality conformance tests requried for each lot of electrodes. On copy shall be forwarded for each lot included in the shipment.

FIGURE 4. Certification of quality conformance inspection. - Continued

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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